

# **Global Strategy: Class #7**

## ***Inclusive Innovation, Climate Change, and Stakeholder Management***

### ***Natura Co.***

**Leandro “Leo” Pongeluppe**

The Wharton School – Management Department

[pongelup@wharton.upenn.edu](mailto:pongelup@wharton.upenn.edu) / [www.LeandroPongeluppe.com](http://www.LeandroPongeluppe.com)



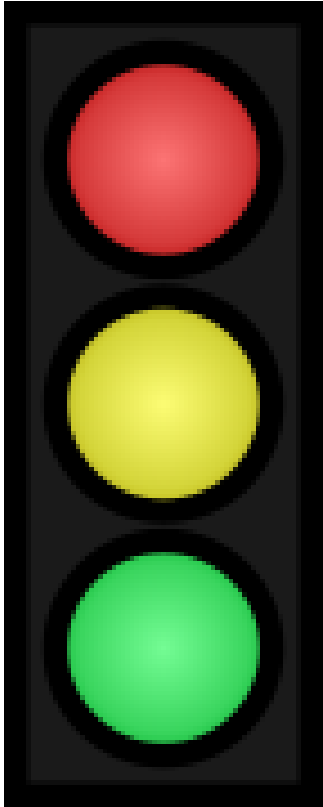
# Last Class

- How should we go? **Transaction Cost Economics (TCE) Framework.**
  - Country risk
    - Political Risk might shift governance curves given the increase in contractual hazards associated with “asset expropriation” risk.
  - Institutional Voids and Business Groups
    - Institutional voids impose difficulties in operations in emerging markets. Firms can act in a countercyclical way.
    - At the same time, institutional voids foster the development of business groups, which act to overcome these voids and reach a competitive position internally and internationally.

## TODAY

- How should we go? **Stakeholder Management Framework**
  - Inclusive Innovation, Climate change, and Stakeholder Management

# What We Are Trying to Solve in the Global Module

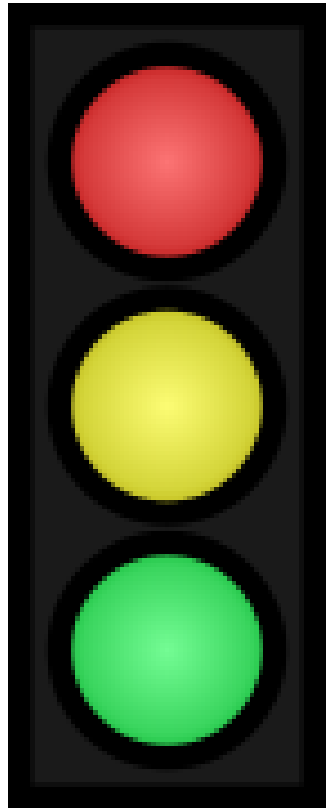


***Should we go abroad?***

***Where should we go?***

***How should we go?***

# What We Are Trying to Solve in the Global Module



*Should we go abroad?*

- **ADDING Value**
- **CIA:** Comparative Institutional Analysis

- Grolsch Beer
- Corruption in International Business

*Where should we go?*

- **CAGE:** Culture, Administrative, Geographic, and Economic
- **AAA:** Adaptation, Aggregation, and Arbitrage

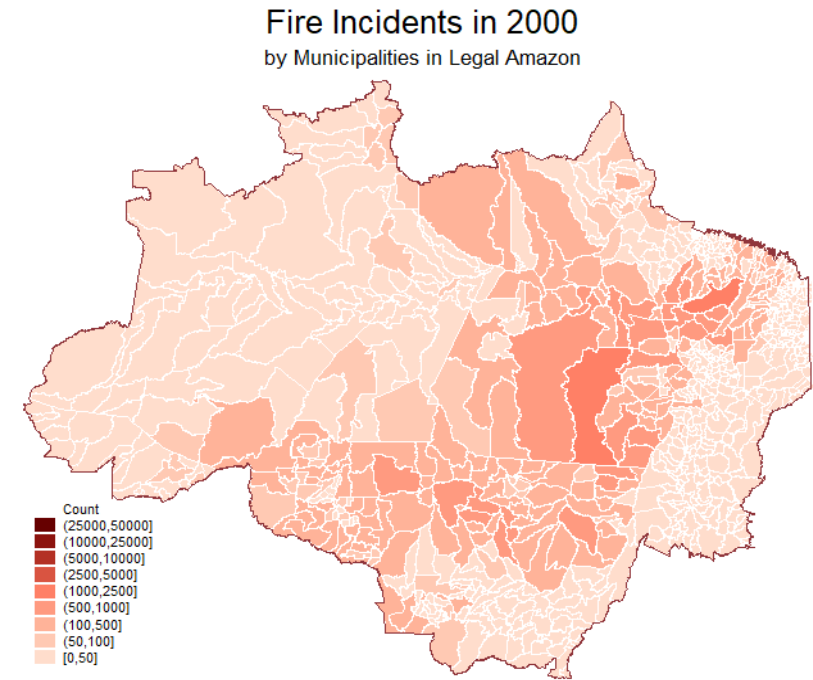
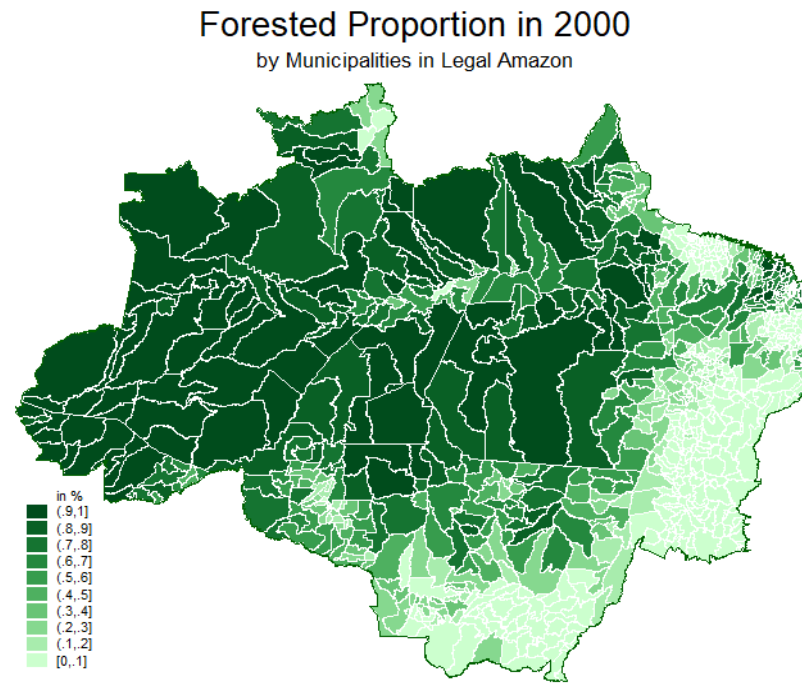
- European Super League
- Apple
- Zara

*How should we go?*

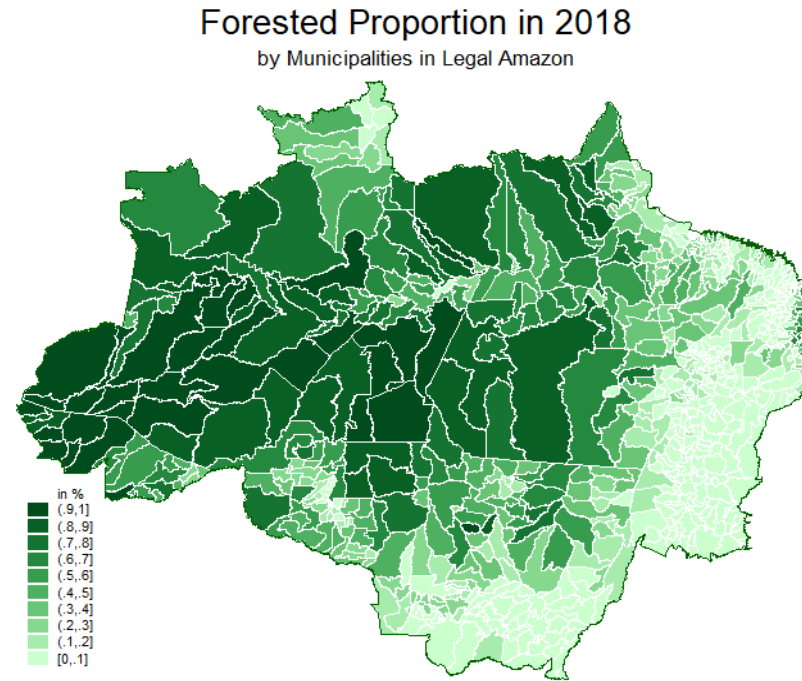
- **TCE:** Transaction Cost Economics
- **SM:** Stakeholders Management

- Walmart in China
- Tata Group
- Natura

# The Forest is Becoming Depleted

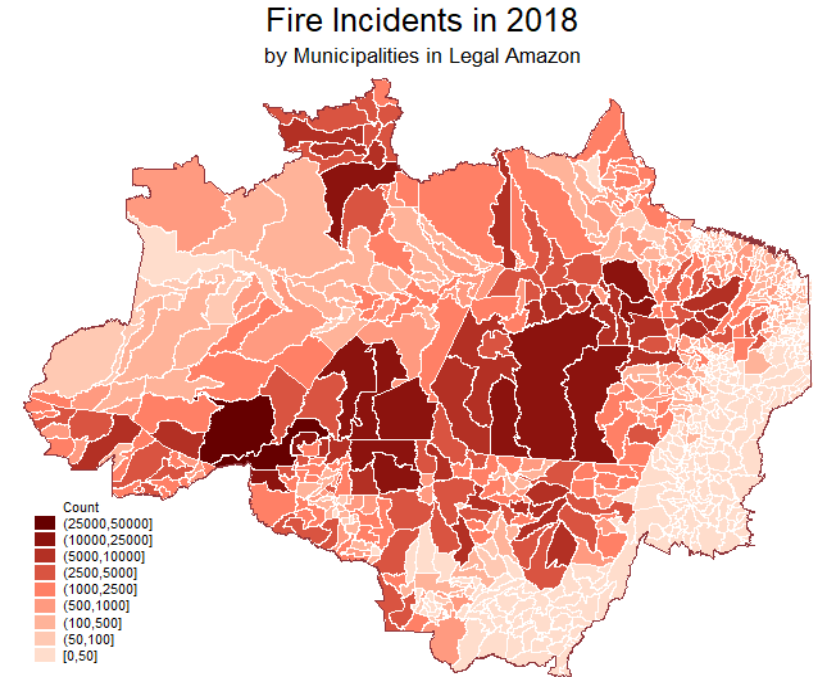


# The Forest is Becoming Depleted



Total forest lost area  
**53.2 million hectares**  
~ **50 million football fields.**

Nota: data from PRODES and IBGE.



Total fire incidents **increase** from  
**60 thousand** to more than  
**900 thousands.**

Nota: data from PRODES and IBGE.

# How can a firm help to save the Amazon rainforest?

---

**Natura Co.** Boehe, Pongeluppe & Lazzarini.  
*“Natura and the development of a sustainable supply chain in the Amazon region.”* Insper Case, 2013.

“Amazon Case Study: Natura’s Supply Chain.”  
*Americas Quarterly*. May 12th, 2021.

Freakonomics Radio, Episode 428 *“The Simple Economics of Saving the Amazon Rain Forest”*.  
Click [here](#) to listen.





[Video Link](#)

# How can a firm help to save the Amazon rainforest?

---

**Natura Co.** Boehe, Pongeluppe & Lazzarini.  
*“Natura and the development of a sustainable supply chain in the Amazon region.”* Insper Case, 2013.



Sources; <https://www.ecohealthalliance.org/2019/08/the-amazon-is-burning>;  
<https://www.dtnext.in/News/World/2019/09/05042942/1175373/Explained-Why-the-Amazon-rainforest-is-on-fire.vpf>

## In-class Group Exercise, Part I

Each group of students will be randomly assigned to the role of the following **stakeholders**:

1. Natura's shareholders
2. Government
3. FASE (grassroots NGO)
4. Local community (Amazonian suppliers)
5. Customers (of cosmetics)

Please discuss in the group (12~15 min) and write one or two paragraphs answering the following question:

**What actions would you require from Natura's managers to guarantee that you, as a stakeholder group, are benefitting from its operations in the Amazon?**

Natura's shareholders

What actions would you require from Natura to guarantee that **YOU**, as a **stakeholder group**, are benefitting from its operations in the Amazon?

Amazonian Supplier



Customers (of cosmetics)

FASE (NGO)

Government

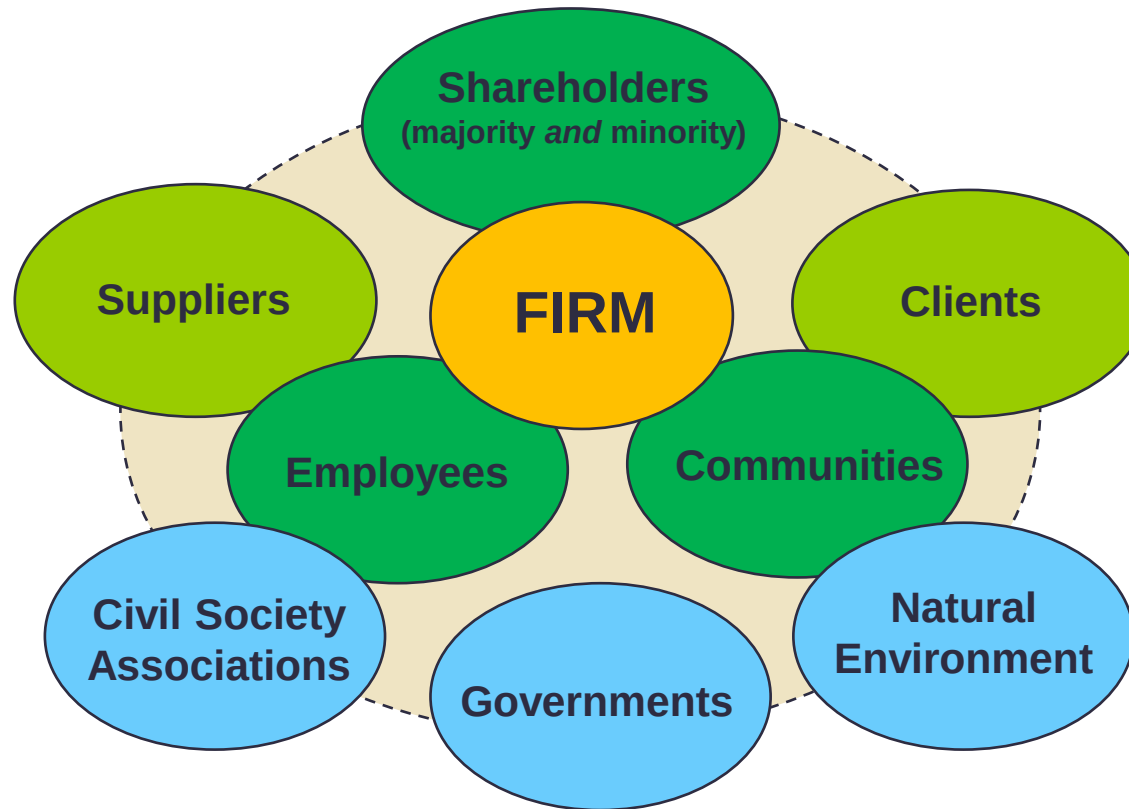
## In-class Group Exercise, Part II

Please discuss in the group (~5 min) and write one paragraph answering the following question:

**To which extent are your demands, as a stakeholder group, consistent with the interests of *another* stakeholder group?**

**(For instance: suppliers can discuss how their demands are in the interests of Natura's shareholders, etc.)**

# The Stakeholder Management Framework



**“Instrumental” stakeholder hypothesis:** benefitting stakeholders or avoiding harm can lead to greater profitability in the long run.

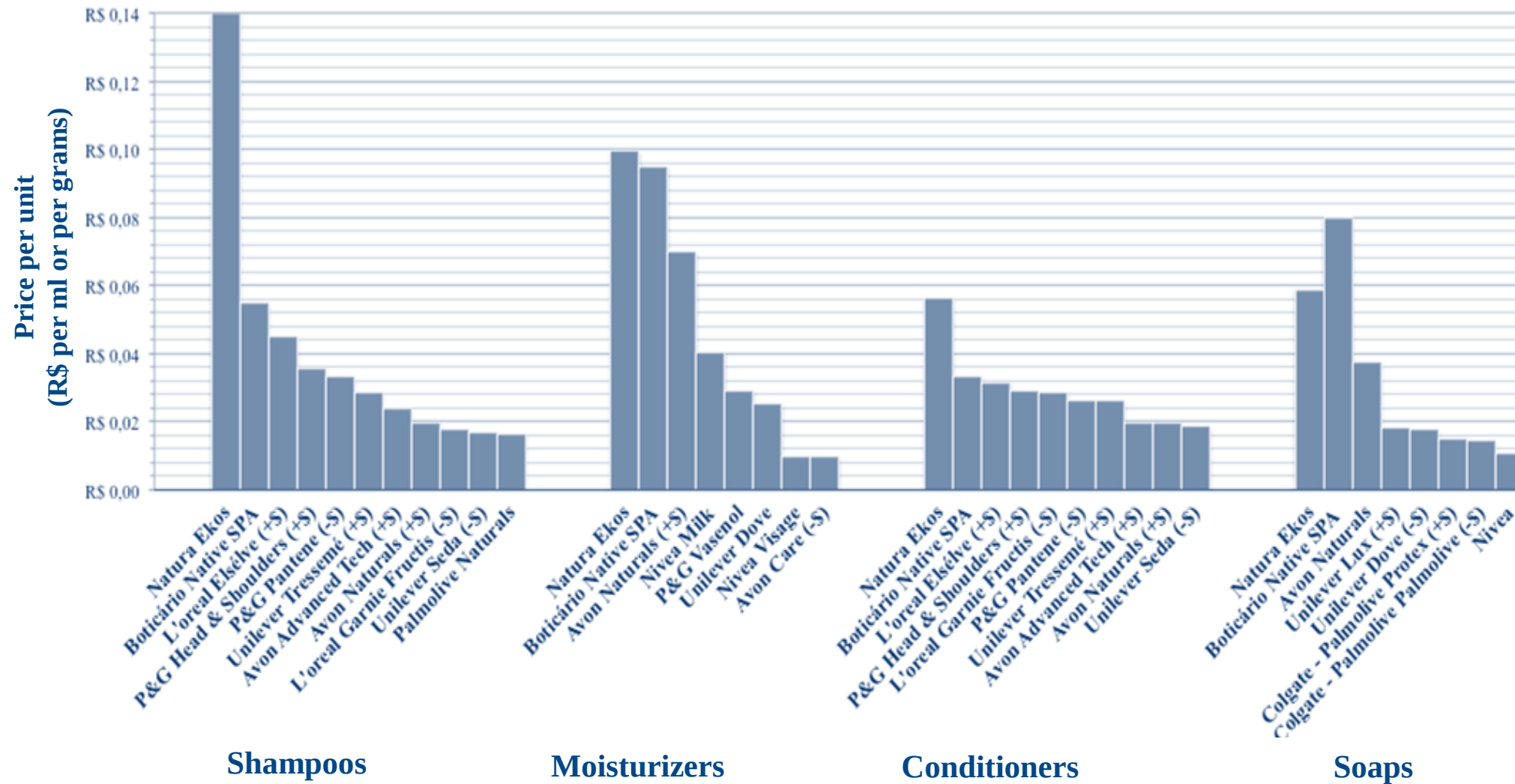
# Natura Co.

---

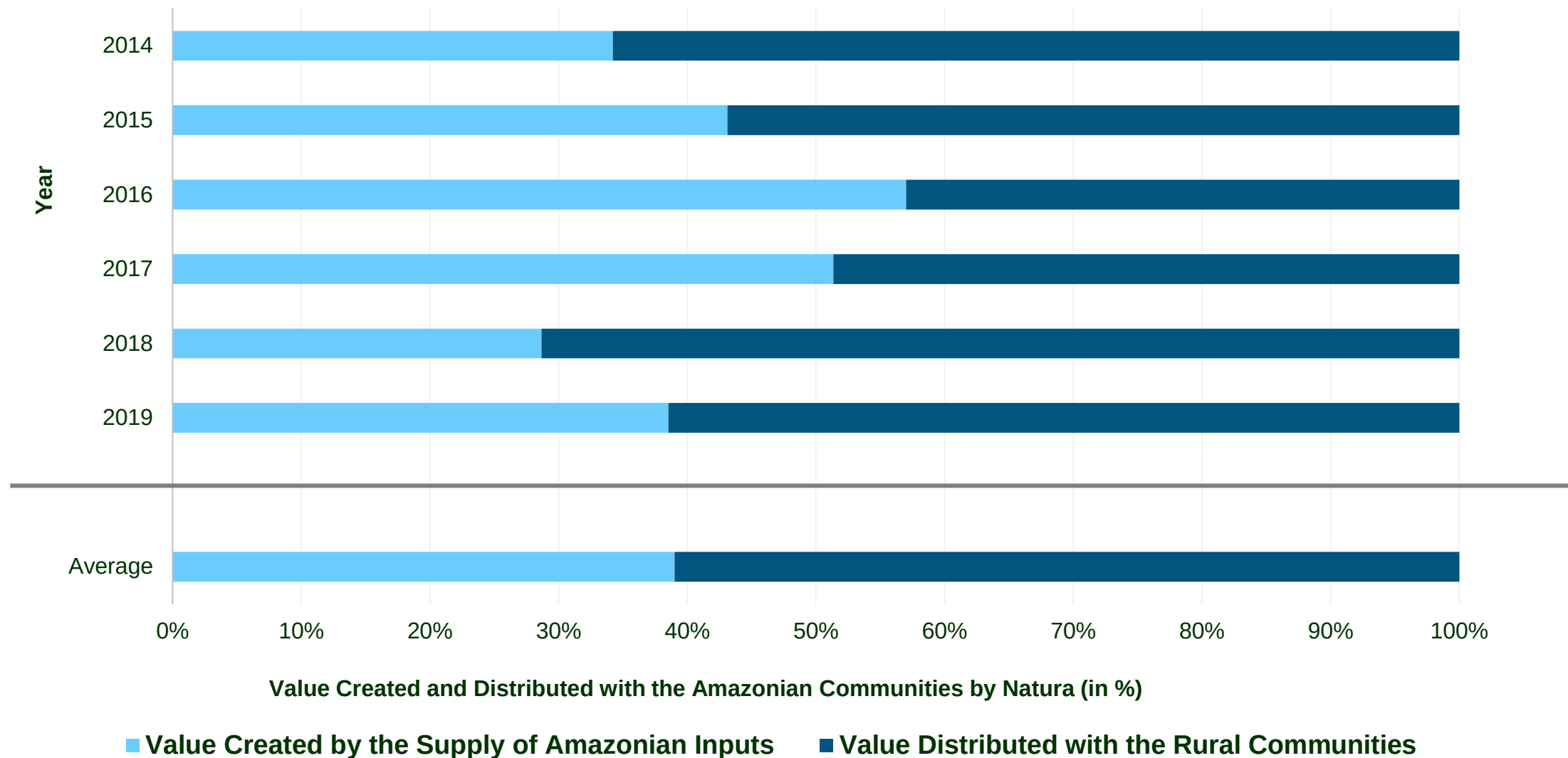
- Founded in 1969 in Brazil. The company operates in **9 countries**, has yearly net revenue of **USD 7 billion (2020)**.
- Internationalization greenfield and through Acquisitions. Natura acquired “**Aesop**” (2014), “**The Body Shop**” (2017), and “**Avon**” (2019).
- **UN Champion of the Earth winner** (2015), the world largest **B-Corp** (Marquis, 2020).
- Natura sells **cosmetics** incorporating 14 different **Amazonian fruits and nuts** (Gatignon & Capron, 2020).



# Natura Ekos commands price premium



# Natura Distributes Value to Align Incentives



Note: Data from Natura's Annual Report 2019 and Annual Report 2016. GRI 203-1 indicator.



[Video Link](#)

**All good, but ...**

**...how do we  
measure the  
impact?**

---



# Satellite Monitoring System

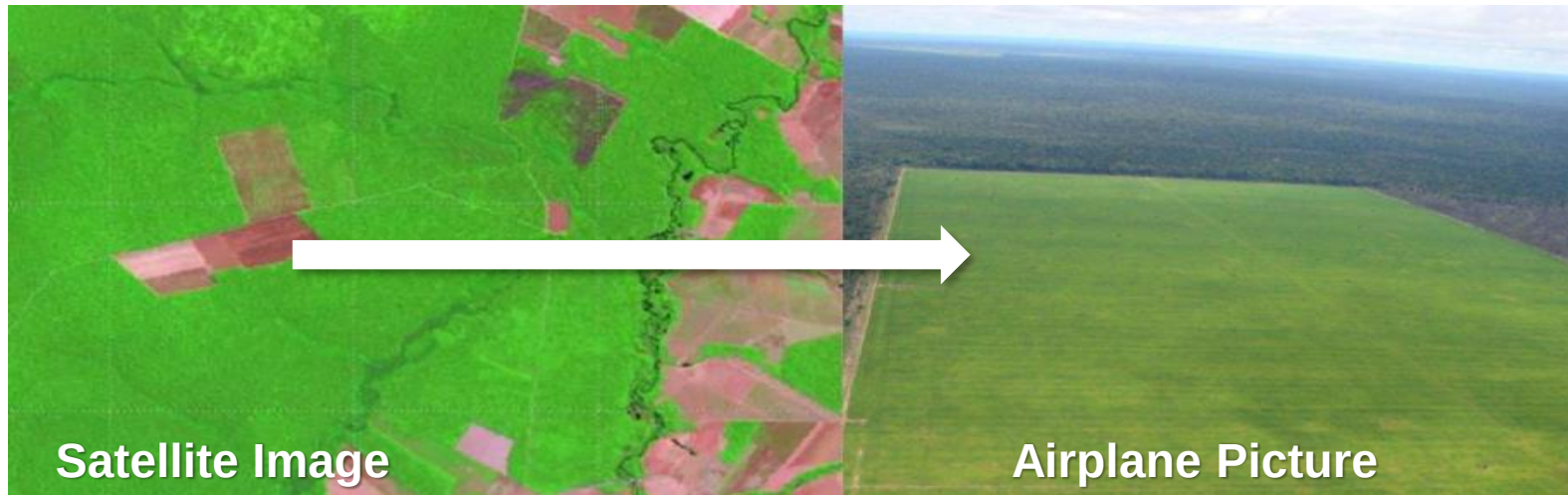
Images **Landsat** (NASA/USA) spatial resolution 30 meters;

Monitoring system has **international reliability** ([Kintisch, 2007- Science](#)).

Detect **deforestation of shallow cut**;

i.e., total deforestation in a window of time;

analyze the yearly increment of deforestation;



Source: PRODES. Available at < [http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes/pdfs/apresentacao\\_prodes-1.pdf](http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes/pdfs/apresentacao_prodes-1.pdf) > And < [http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes/pdfs/Metodologia\\_Prodes\\_Deter\\_revisada.pdf](http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes/pdfs/Metodologia_Prodes_Deter_revisada.pdf) > Accessed on October 22<sup>nd</sup>, 2019

Ano PRODES T



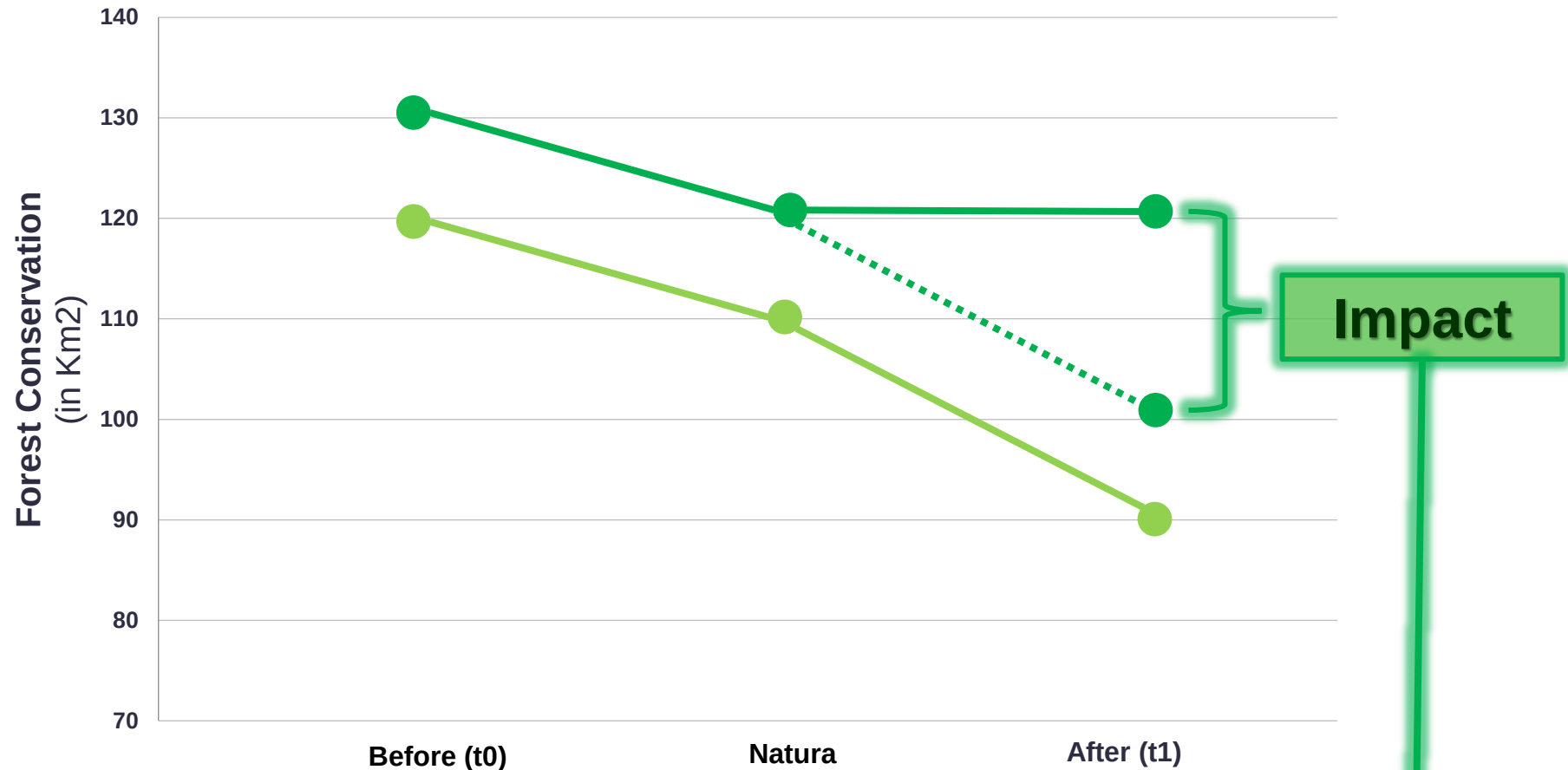
Máscara de todos os desmatamentos até o ano T (inclui as áreas de não floresta)

Ano PRODES T+1



Novos incrementos mapeados no ano T+1

# Additionality Principle



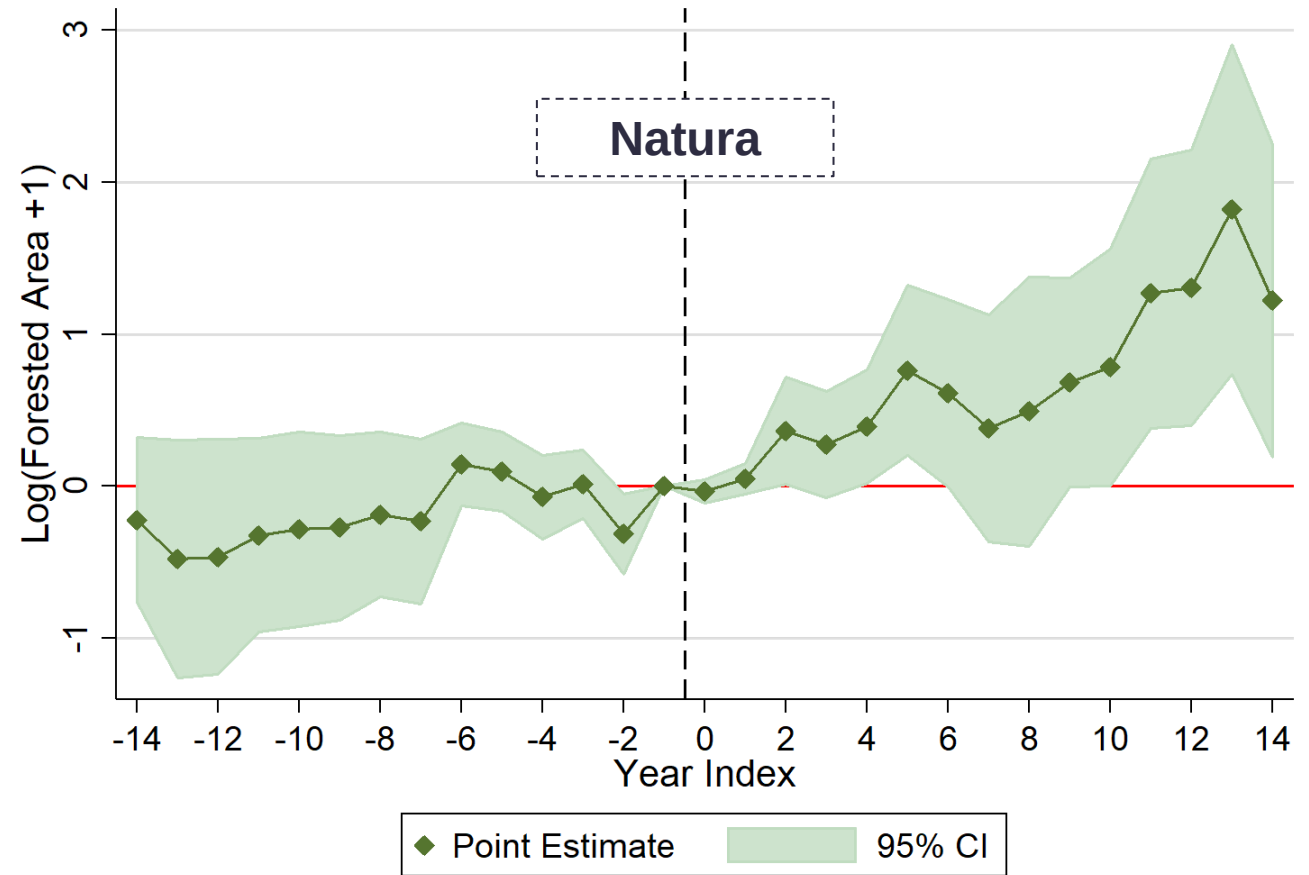
$$\begin{aligned} \text{IMPACT} &= \\ &= [\text{Natura in } t_1 - \text{NonNatura in } t_1] - [\text{Natura in } t_0 - \text{NonNatura in } t_0] \end{aligned}$$

|                                    | (1)                             | (2)                             |
|------------------------------------|---------------------------------|---------------------------------|
| DV = <b>Log(Forested Area + 1)</b> |                                 |                                 |
| <b>Natura</b>                      | <b>0.468*</b><br><b>(0.231)</b> | <b>0.479*</b><br><b>(0.224)</b> |
| Year FE                            | Y                               | Y                               |
| Municipality FE                    | Y                               | Y                               |
| Satellite Controls                 | N                               | Y                               |
| Constant                           | 5.232**<br>(0.0371)             | 5.303**<br>(0.0368)             |
| Observations                       | 14,440                          | 14,440                          |
| R-squared                          | 0.949                           | 0.951                           |
| Number of counties                 | 760                             | 760                             |

Note: Robust standard errors clustered at the municipality level in parentheses.

\*\* p<0.01, \* p<0.05, + p<0.1

## Increased Forest Preservation



Note: standard errors clustered at the municipality level.

On Average, Natura **increases** Forest Preservation by **47.9%** ( $p < 0.05$ )

**1.8 million hectares ~ 1.7 million football fields**

**~ 145 million tons of Carbon.**

# Estimation of Natura's Effect in Carbon Tons

Estimated Effect of Natura in Carbon Tons saved and potential Price of the Positive Externality

| Carbon stock in Amazon (tC/ha) | - 2sd | - 1sd | Mean | + 1sd | + 2sd |
|--------------------------------|-------|-------|------|-------|-------|
|                                | 26.4  | 53.5  | 80.6 | 107.7 | 134.8 |

Note 1: Carbon stock in Amazon considers the difference between forested and deforested areas. Source (Souza-Rodrigues, 2018: 2730).

Note 2: Natura Estimated conservation takes into account the total number of hectares “saved” by Natura (total of 1.807.640 hectares), which are based on the estimations presented in the previous slide.

Note 3: Carbon prices set considering historical prices and the price on January 1st, 2020, which equals € 24.32/tC. Source: Bayer and Aklin (2020) and ICE EUA Futures. Available at: <<https://www.theice.com/products/197/EUA-Futures/data?marketId=5115271&span=3>> Accessed on April 16th, 2020.

# Estimation of Natura's Effect in Carbon Tons

## Estimated Effect of Natura in Carbon Tons saved and potential Price of the Positive Externality

| Carbon stock in Amazon (tC/ha)                                           |      | - 2sd                                            | - 1sd      | Mean        | + 1sd       | + 2sd       |
|--------------------------------------------------------------------------|------|--------------------------------------------------|------------|-------------|-------------|-------------|
|                                                                          |      | 26.4                                             | 53.5       | 80.6        | 107.7       | 134.8       |
| Natura Estimated Conservation in Amazon (in tC)                          |      | 47,721,702                                       | 96,708,752 | 145,695,802 | 194,682,852 | 243,669,903 |
|                                                                          |      | Natura Estimated Conservation in (millions €/tC) |            |             |             |             |
| Price of Carbon under the European Union Emissions Trading System (€/tC) | € 5  | € 239                                            | € 484      | € 728       | € 973       | € 1,218     |
|                                                                          | € 15 | € 716                                            | € 1,451    | € 2,185     | € 2,920     | € 3,655     |
|                                                                          | € 25 | € 1,193                                          | € 2,418    | € 3,642     | € 4,867     | € 6,092     |
|                                                                          | € 35 | € 1,670                                          | € 3,385    | € 5,099     | € 6,814     | € 8,528     |

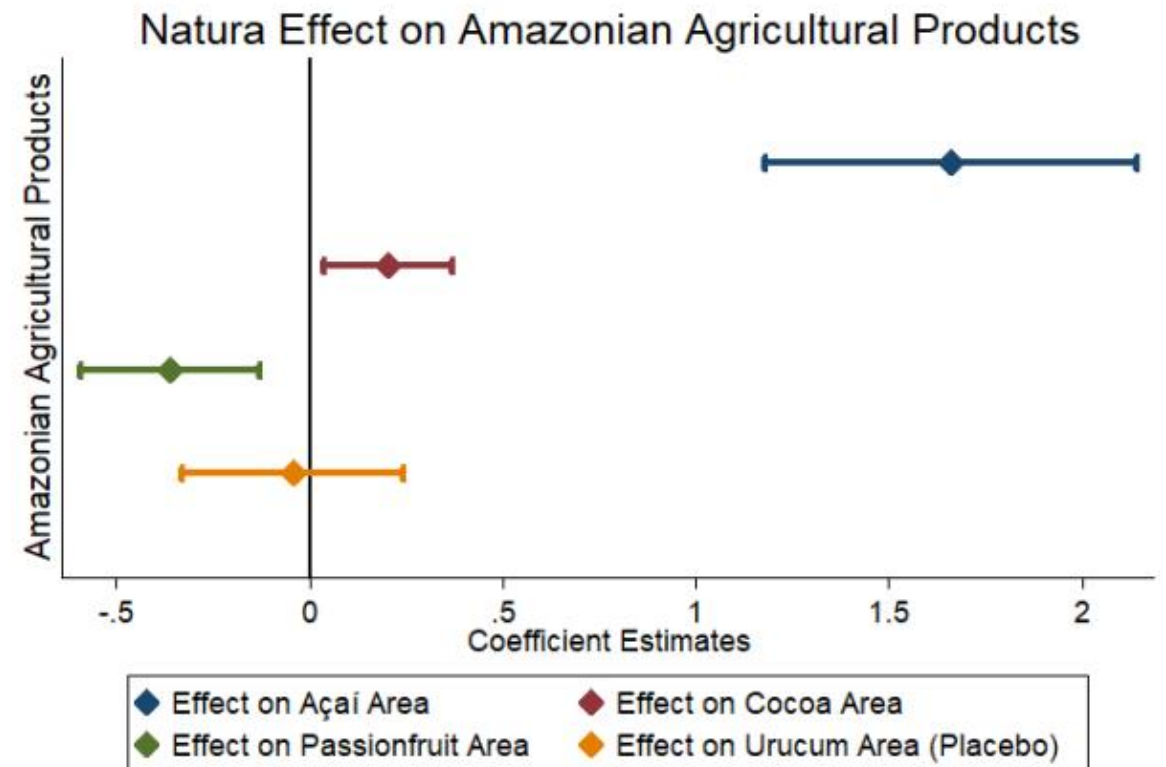
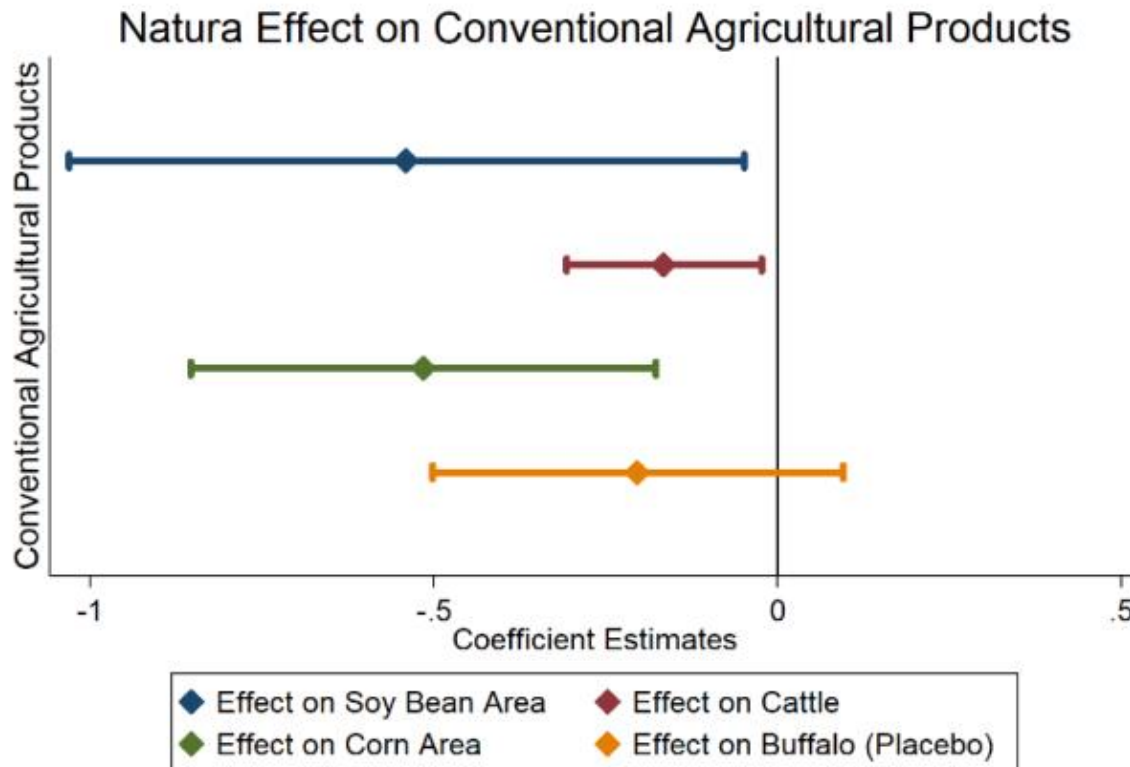
Note 1: Carbon stock in Amazon considers the difference between forested and deforested areas. Source (Souza-Rodrigues, 2018: 2730).

Note 2: Natura Estimated conservation takes into account the total number of hectares "saved" by Natura (total of 1.807.640 hectares), which are based on the estimations presented in the previous slide.

Note 3: Carbon prices set considering historical prices and the price on January 1st, 2020, which equals € 24.32/tC. Source: Bayer and Aklin (2020) and ICE EUA Futures. Available at: <<https://www.theice.com/products/197/EUA-Futures/data?marketId=5115271&span=3>> Accessed on April 16th, 2020.

# Conventional vs. Amazonian Agriculture

- Reduction in Conventional Agricultural Products, but an Increase in Amazonian Agricultural Products
- This leads to the increase in **Forested Areas**



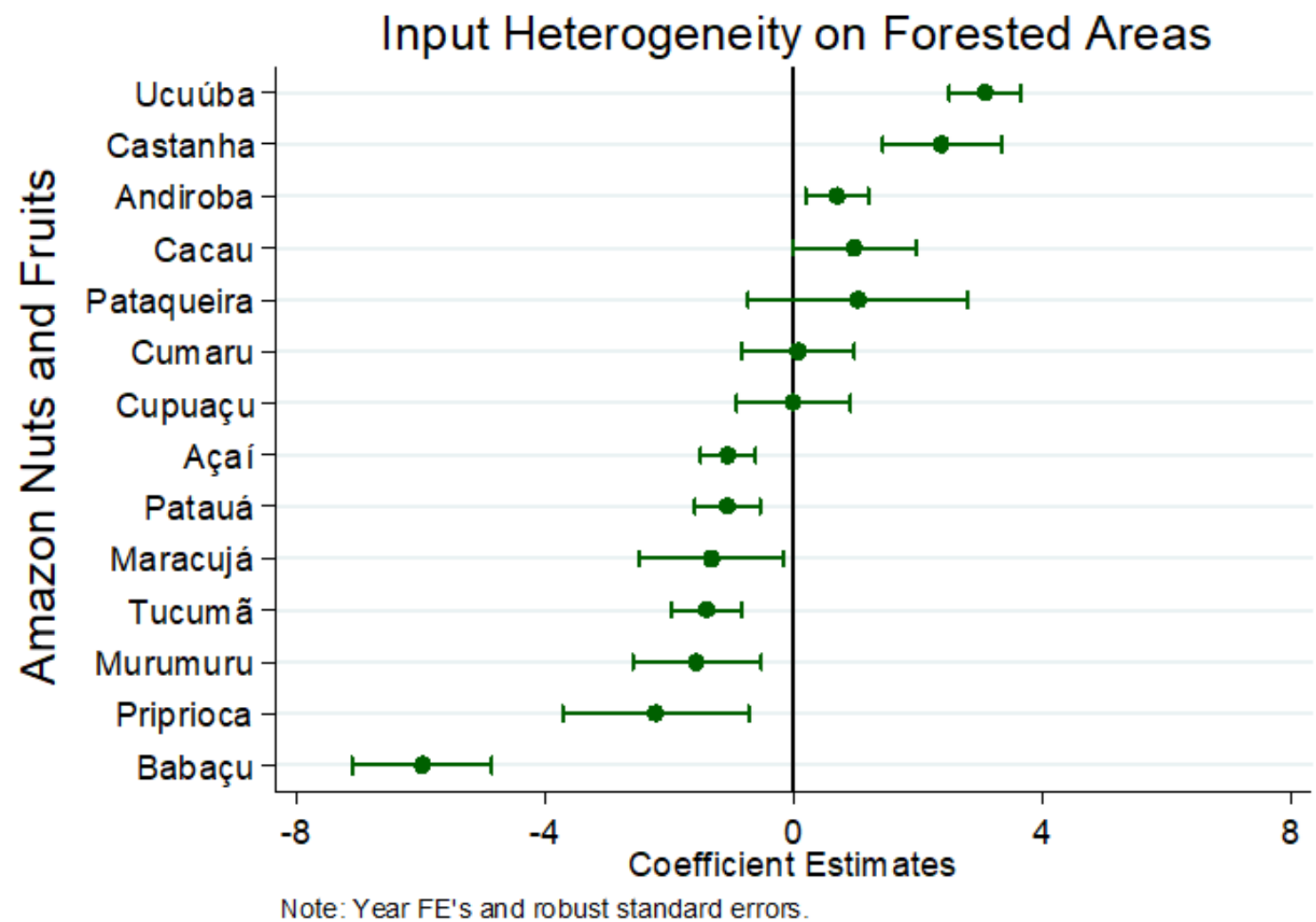
## The power of incentive alignment with stakeholders

***“If it wasn't for cocoa, you had to move cattle (...) with cocoa, a family that has 10 hectares of cocoa already supports themselves (...) this 10 hectares generate about R\$3.000 month.”***

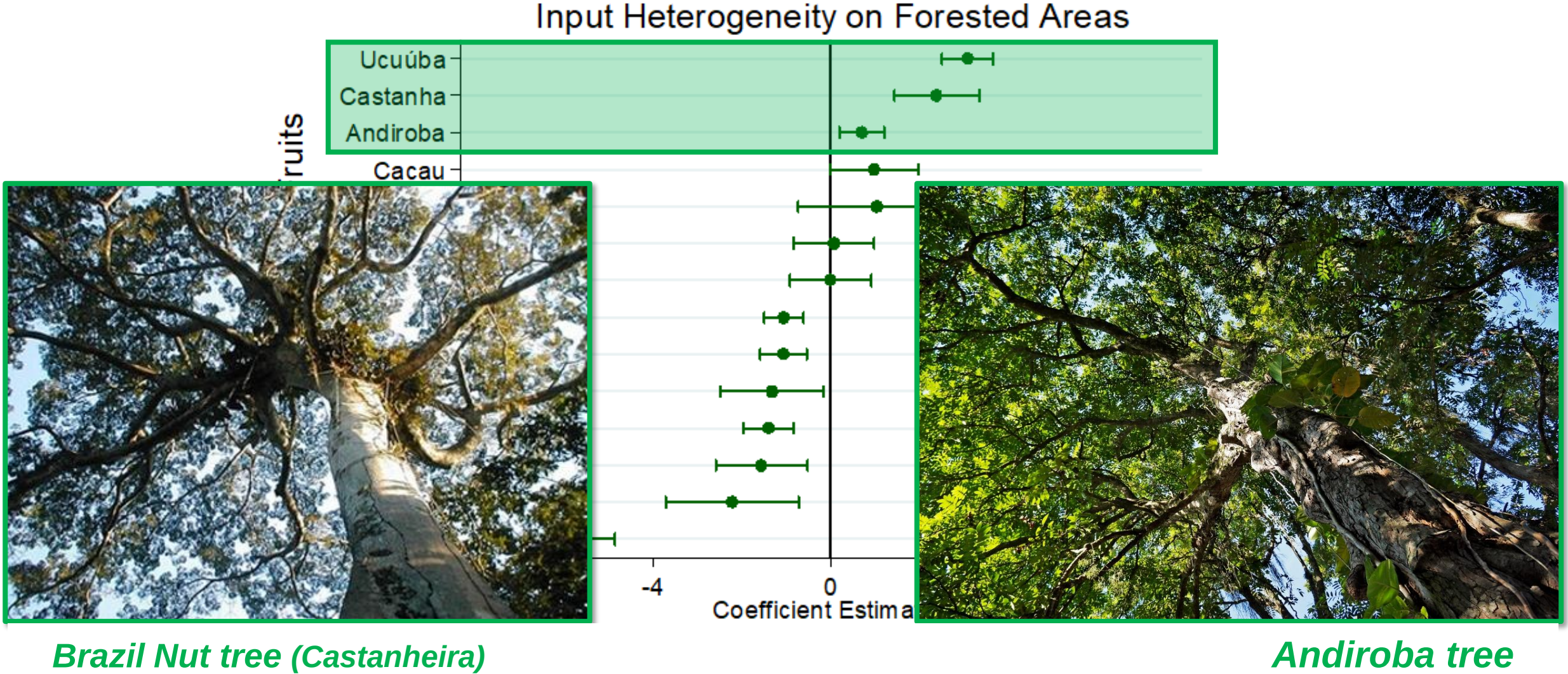
*-- Reginaldo Silva, organic cocoa supplier.*



# Effects on Forest Conservation and Regeneration

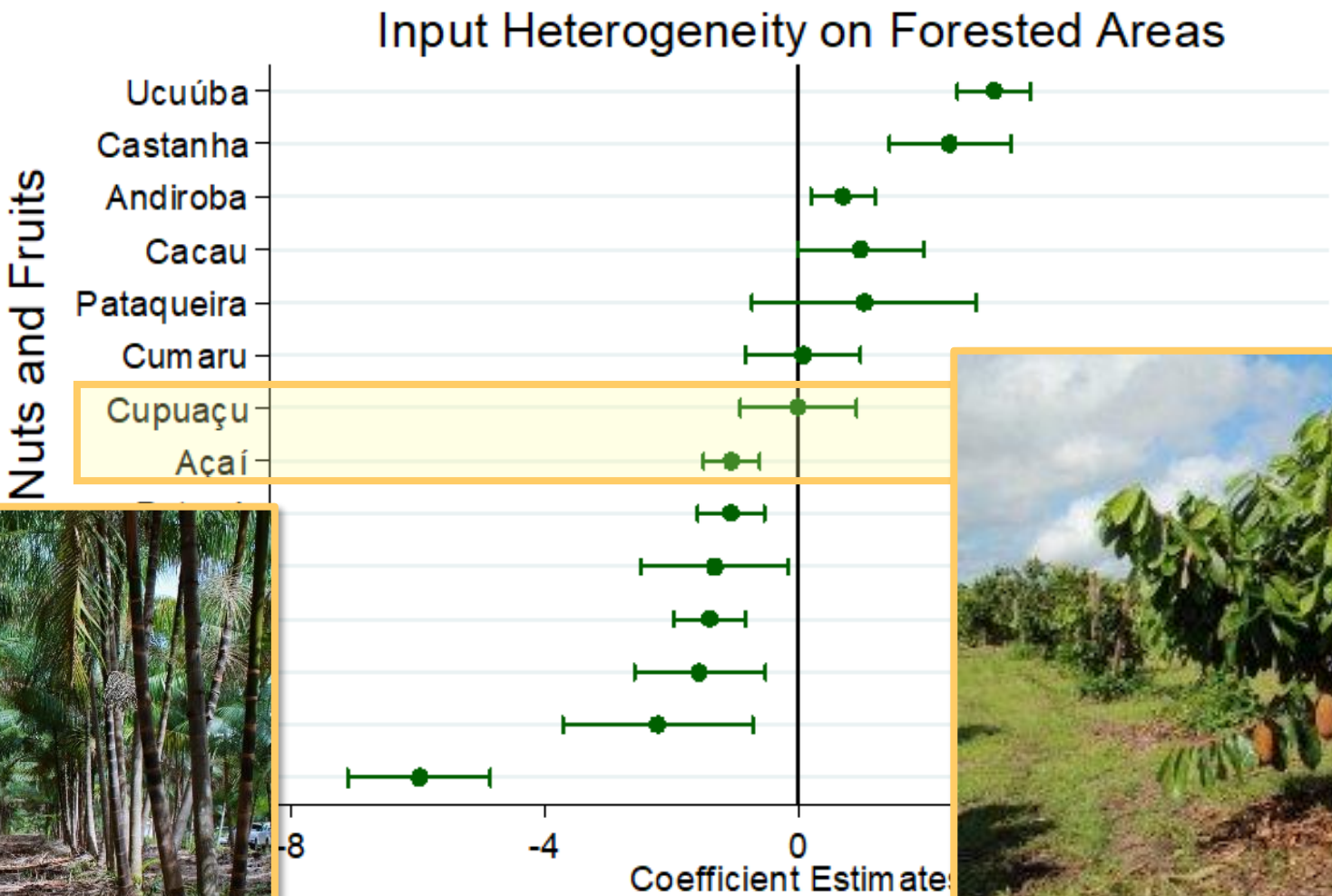


# Effects on Forest Conservation and Regeneration



# Effects on Forest Conservation and Regeneration

Açaí palm

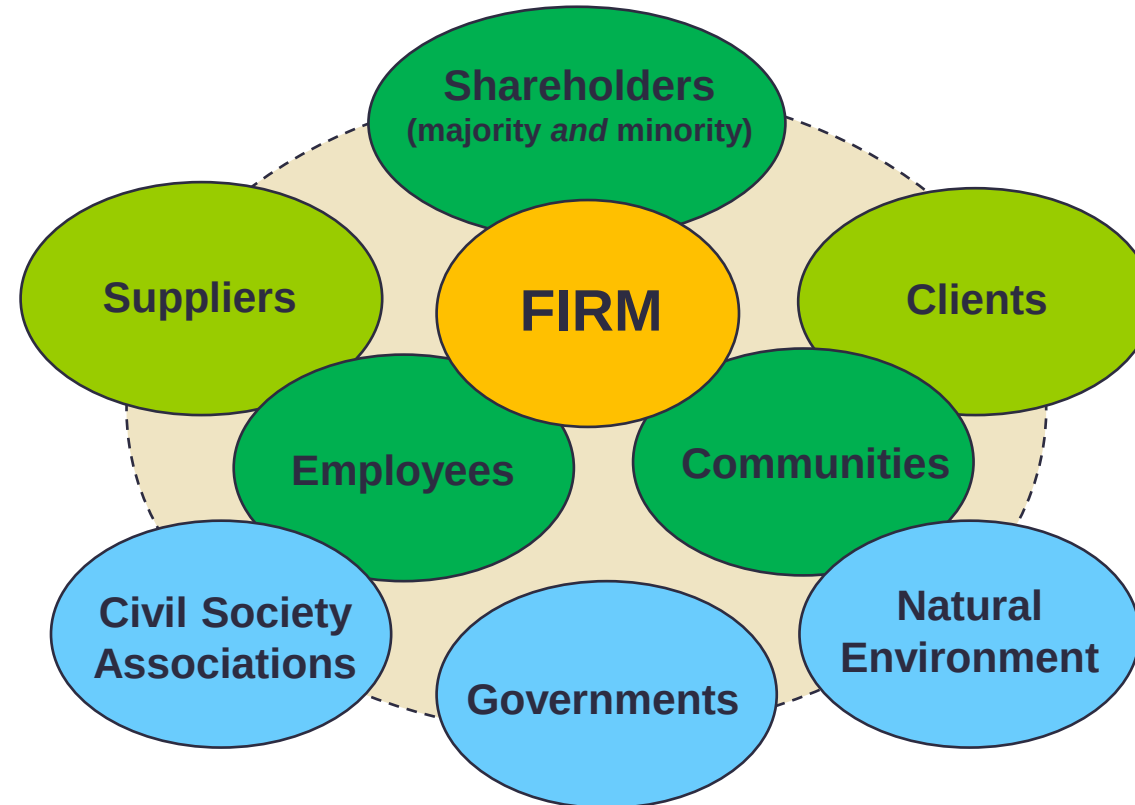


Cupuaçu trees



Note: Year FE's and robust standard errors.

# The Stakeholder Management Framework



**BUT, sometimes firms might want to mislead shareholders and other stakeholders**

**“Greenwashing”**

# A new toolkit to measure “greenwashing”

- **Textual Analysis** (using annual/sustainability reports)

- Why? There is evidence that firms use language to obfuscate environmental performance mandated by standardized reporting
- We create a dictionary based on the “Corporate Social Responsibility Words-Environment” dictionary (CAT Scanner)
- We expand this “Envirolex” dictionary. Now it contains 321 terms, including Amazon specific ones (e.g., Amazon biome, Amazon rainforest, etc.)
- We count the frequency of Envirolex words

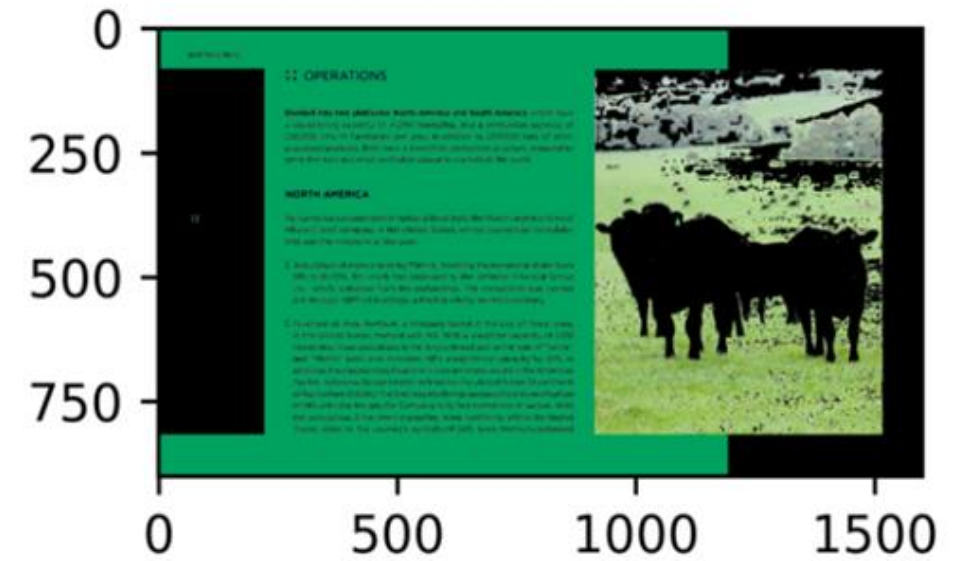
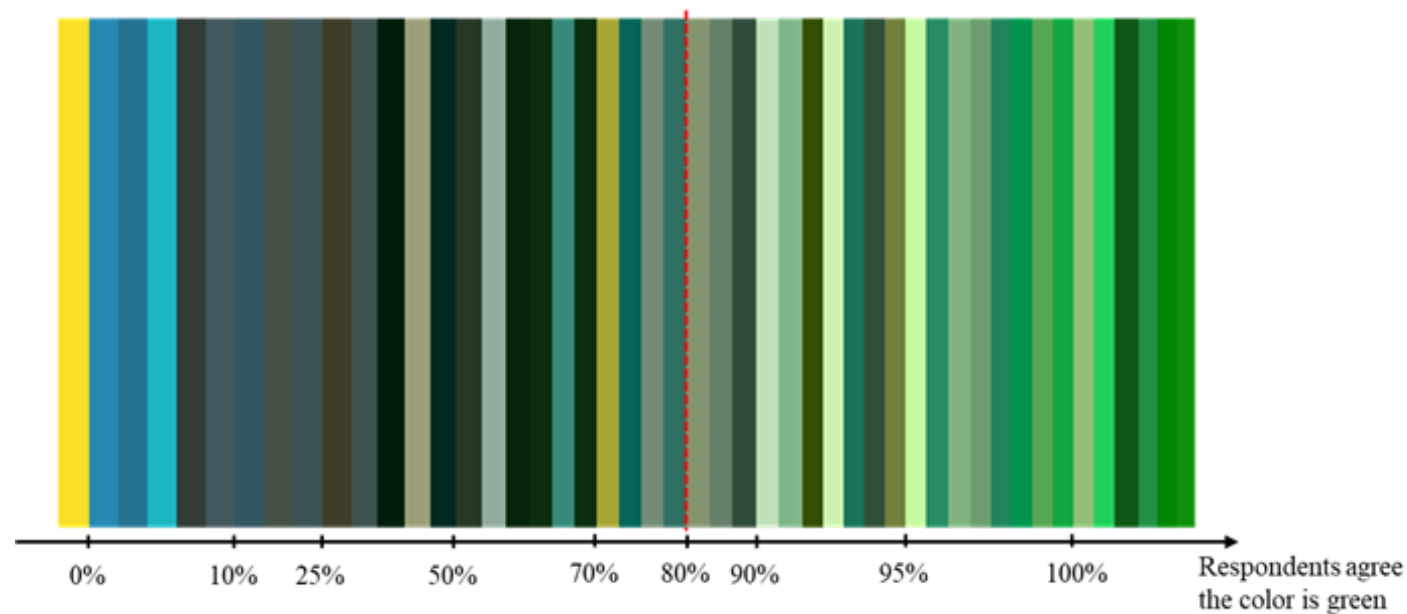
- **Sentiment Analysis**

- Use the Loughran and McDonald (L&M) dictionary (H4N) to measure sentiment
- We count positive/negative terms in the report and in sentences that have Envirolex words

- **Visual Analysis** (image analysis to figure out how green a report is)

# Visual analysis

- We developed a **Python algorithm** to identify the number of **green pixels** in a firm's report
- **But, what is green?**
  - MTurk survey with 1,000 respondents helped us to define what the human eye considers green.
  - Define green using max/min boundaries of the HSV (Hue, Saturation, Value) space that comes out of our survey.



# Results

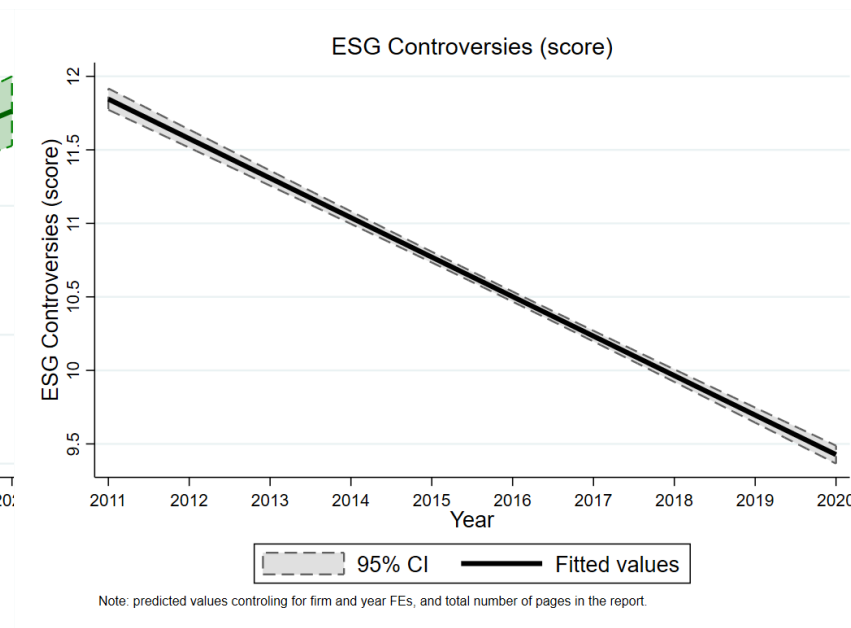
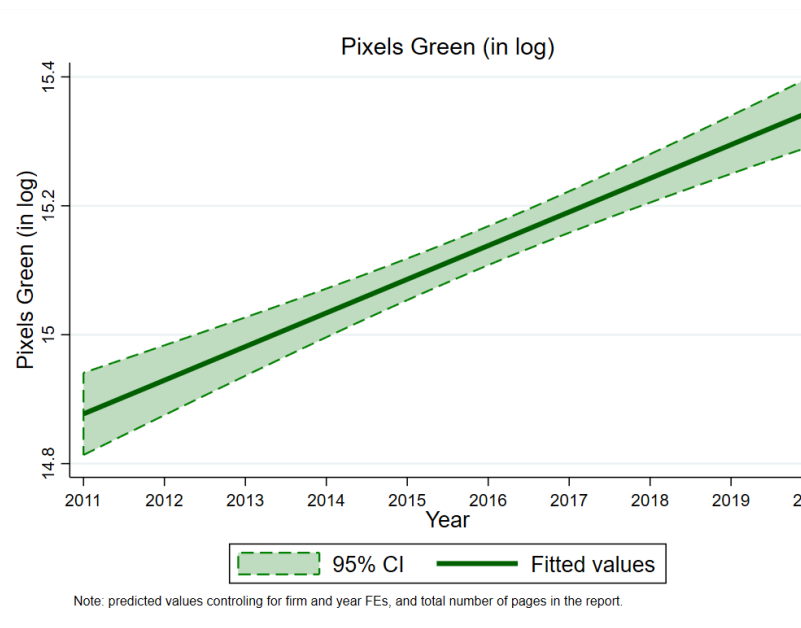
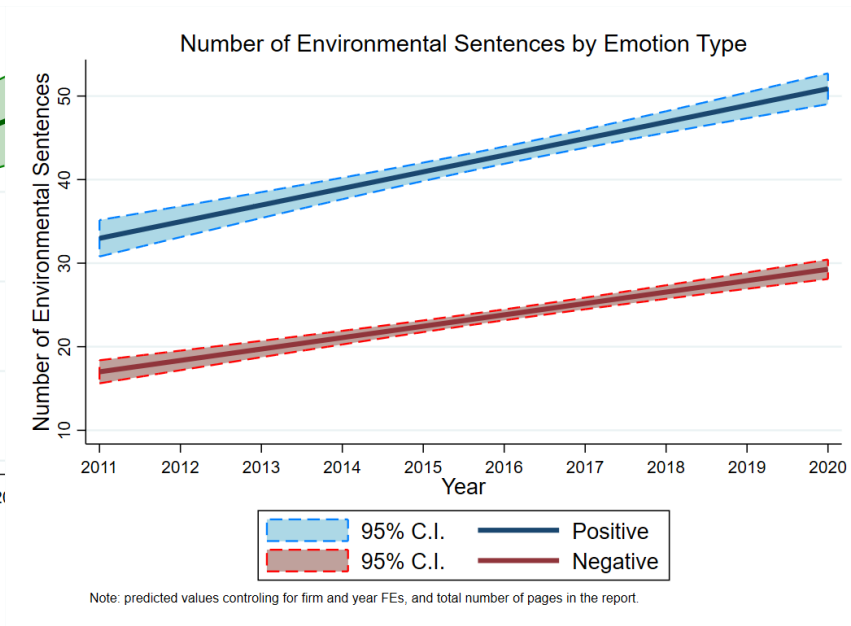
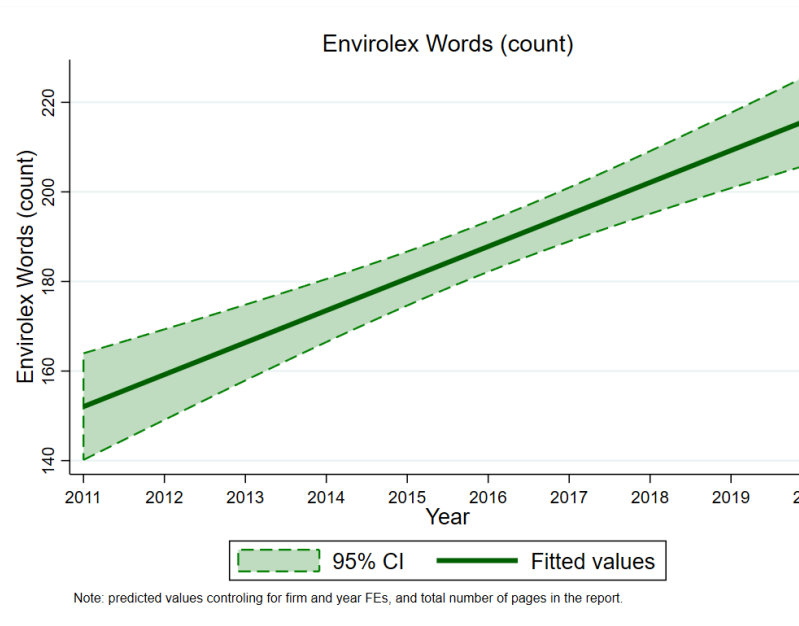
## Over ten years

- 73 more environment-related words ( $p = 0.000$ )
- 16.68 additional environmental sentences with positive sentiment ( $p = 0.000$ )
- 14.52 additional environmental sentences with negative sentiment ( $p = 0.000$ )
- 36% higher number of green pixels ( $p = 0.026$ ) ~ 2.75 million more green pixels

## BUT...

- 2 points lower ESG Controversies score. That is, more fines related to ESG wrongdoing ( $p = 0.000$ )

Note: ESG controversies score from Thomson-Refinitiv database rates firms according to the number of environmental penalties and wrongdoings the firm has in a year.





## Can MNCs help solve global socioenvironmental problems?

---

**Narsalay, Pongeluppe, & Light (2015)**  
“The Hidden Pitfalls of Inclusive Innovation,”  
*Stanford Social Innovation Review*. Winter:  
48-53

# “Inclusive Innovation” Project

- One billion poor people from emerging markets will rise to the middle class in the next several decades → Huge business opportunity
- Partnership among:
  - Accenture Institute for High Performance
  - Insper (São Paulo, Brazil)
  - Lagos Business School (Lagos, Nigeria)
  - South Asia Initiative at Harvard University (Cambridge, MA, USA)

**What are the drivers for MNCs to successfully engage with Bottom-of-the-Pyramid (BoP) populations, generating “Inclusive Innovations”?**

# Firms and locations

| Country                                                                             | Agriculture          | Automobiles                | Consumer Goods                   | Mining | Medical Devices | Telecomm. | Banking  |
|-------------------------------------------------------------------------------------|----------------------|----------------------------|----------------------------------|--------|-----------------|-----------|----------|
|    | UNICA<br>&<br>Klabin |                            | Natura                           | VALE   |                 |           | Itaú     |
|    |                      | Shanghai-GM<br>&<br>Vuling | Alibaba<br>&<br>Haier            |        | Siemens         | Nokia     |          |
|    | ESOKO                |                            | Promasidor<br>&<br>Nestle        |        |                 |           |          |
|  | ITC                  | Tata Motors                | Hindustan<br>Unilever<br>Limited |        |                 | Idea      | Yes Bank |

# Inclusive Business Innovations (IBIs) Pitfalls

- Lack of buy-in from top managers – “one strike out”
  - Ex.: **Hindustan Unilever Limited (HUL)** (India) directors defending the idea.
- Focus on the wrong performance metrics
  - Ex.: **ESOKO** (Ghana) creating the infrastructure and training SMS about prices on agricultural goods.
- Failing to recruit talented executives
- Partnering with wrong organizations
- Using old business models
  - Ex.: **Natura** (Brazil) instead works with the communities to create new products and business models.



Hindustan Unilever Limited

esoko



natura

***“The SDGs offer the potential to unite the most powerful organizations in the world – governments, corporations, and civil society – behind a single agenda to save humanity and the planet from suffering and devastation.***

***Including the private sector in their development and execution can greatly magnify the odds of success.”***

## SUSTAINABLE DEVELOPMENT GOALS



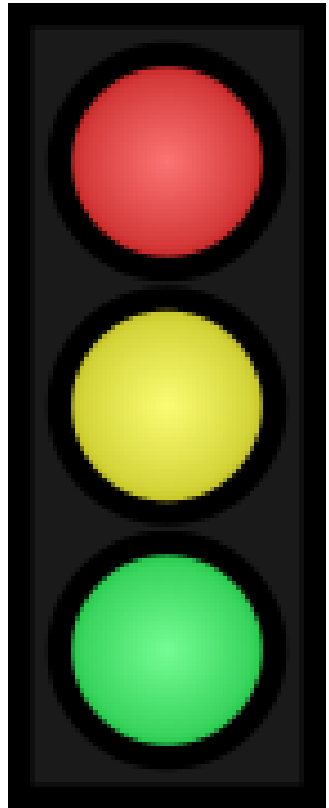
[This Photo](#) by Unknown Author is licensed under [CC BY](#)

(Kramer, Agarwal, & Srinivas, 2019: 9)

# Key Takeaways

- How should we go? **Stakeholder Management Framework**
  - An adequate stakeholder management can be an effective way by which firms (MNCs) could contribute to solve many global problems.
  - The internalization of externalities is a clever way by which firms can align economic, social, and ecological incentives.
  - However, some firms might be strategic in their ESG disclosure aiming to mislead their shareholders and other stakeholders.
  - The additionality principle enables us to measure potential socioenvironmental impact and verify whether the firm is causing this impact or not.
  - The UN-SDGs provide a roadmap on which goals firms might contribute to achieve by 2030.

# Wrapping up the Global module...



*Should we go abroad?*

- **ADDING Value**
- **CIA:** Comparative Institutional Analysis

- Grolsch Beer
- Corruption in International Business

*Where should we go?*

- **CAGE:** Culture, Administrative, Geographic, and Economic
- **AAA:** Adaptation, Aggregation, and Arbitrage

- European Super League
- Apple
- Zara

*How should we go?*

- **TCE:** Transaction Cost Economics
- **SM:** Stakeholders Management

- Walmart in China
- Tata Group
- Natura



# Looking Ahead: Group Presentations

---

- **Next Class:**
  - Class Presentations: Monday and Wednesday
  - Upload slides until December 5<sup>th</sup> by 8am
  - Final paper due December 8<sup>th</sup> by noon

**Thank you.  
Please stay in touch!  
Find me on LinkedIn or Email.**

**Leo Pongeluppe**  
[pongelup@wharton.upenn.edu](mailto:pongelup@wharton.upenn.edu)

